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Forecasting Indonesian Coal Stock Prices Using a Hybrid ARIMA-LSTM Model in a Web-Based Information System

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Abstract. Coal-sector stock prices are highly volatile and are influenced by changing market conditions, requiring forecasting methods capable of capturing linear and nonlinear patterns simultaneously. This study develops a hybrid Autoregressive Integrated Moving Average-Long Short-Term Memory (ARIMA-LSTM) framework for the daily closing prices of ADRO, PTBA, INDY, and ITMG obtained from Yahoo Finance through December 2024. ARIMA models the linear component, whereas LSTM learns the nonlinear structure remaining in the ARIMA residuals. Out-of-sample evaluation was conducted for January 2023-December 2024 using MAE, RMSE, MAPE, and R^2. The model produced different levels of accuracy across stocks: PTBA achieved the lowest MAPE at 5.29%, followed by ITMG at 8.50%, ADRO at 11.98%, and INDY at 29.94%. Positive R^2 values were obtained for ADRO and PTBA, while negative values for ITMG and INDY indicate that the same hybrid configuration did not generalize equally across all stocks. The selected models were subsequently used to generate exploratory monthly projections for 2025-2045 and present them through an interactive web dashboard. The study contributes a stock-specific evaluation of hybrid forecasting and an accessible analytical interface for monitoring long-term coal-sector price scenarios.

Keywords: ARIMA-LSTM, coal stocks, forecasting, machine learning, interactive dashboard

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1. INTRODUCTION

Exports are one of the main pillars of international trade and play a strategic role in driving national economic growth through the utilization of natural resources, expertise, and comparative advantages [1]. Indonesia plays an important role in the energy sector, contributing significantly to the national economy through coal, natural gas, and new and renewable energy commodities. Indonesia is one of the world's largest coal producers and exporters, with reserves of 38.84 billion tons and annual production of approximately 600 million tons [2]. Coal contributes around 5-8% of national gross domestic product and dominates approximately 80% of the mining sector's contribution, making it one of Indonesia's leading export commodities alongside palm oil [3], [4].

The extensive development of coal-production infrastructure and strong demand from Asian and European markets have strengthened Indonesia's position in the global energy market [5]. Indonesian coal is internationally competitive because of its relatively low sulfur content and high calorific value [6]. In the stock market, coal-sector companies also attract investor attention because their performance is closely associated with reference coal prices, export demand, and national economic conditions [7]. Nevertheless, the global commitment to sustainable development and the transition toward a low-carbon economy exposes the coal industry to environmental and market challenges, particularly those associated with production waste and changing energy policies [8].

From a circular-economy perspective, coal waste can be utilized as an alternative energy product, including coal-waste briquettes. This approach may improve resource efficiency, diversify energy products, and create additional economic value and export opportunities, especially for developing countries that remain dependent on solid fuels [9]. Consequently, the development of waste-based coal products may contribute to more sustainable environmental management while supporting economic activity in the energy sector [10].

As modern financial systems develop, the stock market has become an important indicator of investor expectations regarding future economic and industrial performance. Coal-sector stock prices can therefore be used to examine market perceptions of the industry's prospects. However, digital access to financial markets does not eliminate the risk of price fluctuations caused by changes in supply, demand, policy, and global commodity conditions [11]. Statistics and machine-learning methods are increasingly used

to model such dynamics because they can learn patterns from historical prices and relevant economic indicators [12].

Previous research comparing Support Vector Regression, Random Forest, and K-Nearest Neighbors for Indonesian stock-market prediction found that Support Vector Regression produced the lowest RMSE and MAE for the selected IDX30 stocks. Its performance was attributed to its ability to model nonlinear relationships through appropriate kernel functions, while the other models showed weaker performance under rapid price changes [13]. Hybrid forecasting has also been proposed to combine the interpretable linear structure captured by statistical models with nonlinear patterns learned by machine-learning models [14].

Most previous studies have focused primarily on improving prediction accuracy without directly connecting sector-specific forecasts to long-term economic and analytical needs. Studies integrating Indonesian coal-sector stocks, stock-specific out-of-sample evaluation, long-term projections to 2045, and an interactive dashboard remain limited. Therefore, this study develops and evaluates a hybrid ARIMA-LSTM framework for four Indonesian coal stocks, compares model performance across companies, generates exploratory projections for 2025-2045, and presents the results through a web-based dashboard. The contribution lies in demonstrating that hybrid-model performance must be assessed separately for stocks with different price dynamics rather than assumed to be uniformly superior.

2. METHODS

2.1. Research Framework

This study applies a structured forecasting and information-system workflow. Daily closing-price data are acquired and checked for missing or duplicate observations before being divided chronologically into training and testing periods. The complete pipeline is then executed independently for ADRO, PTBA, INDY, and ITMG. For each company, ARIMA captures the linear component, while an LSTM network models nonlinear temporal patterns. In the hybrid model, LSTM is trained on the ARIMA residuals and its predicted residual is added to the ARIMA forecast. The resulting forecasts are evaluated using MAE, RMSE, MAPE, and R^2, after which the selected company-specific models generate monthly projections for 2025-2045 that are presented through a web-based interactive dashboard.

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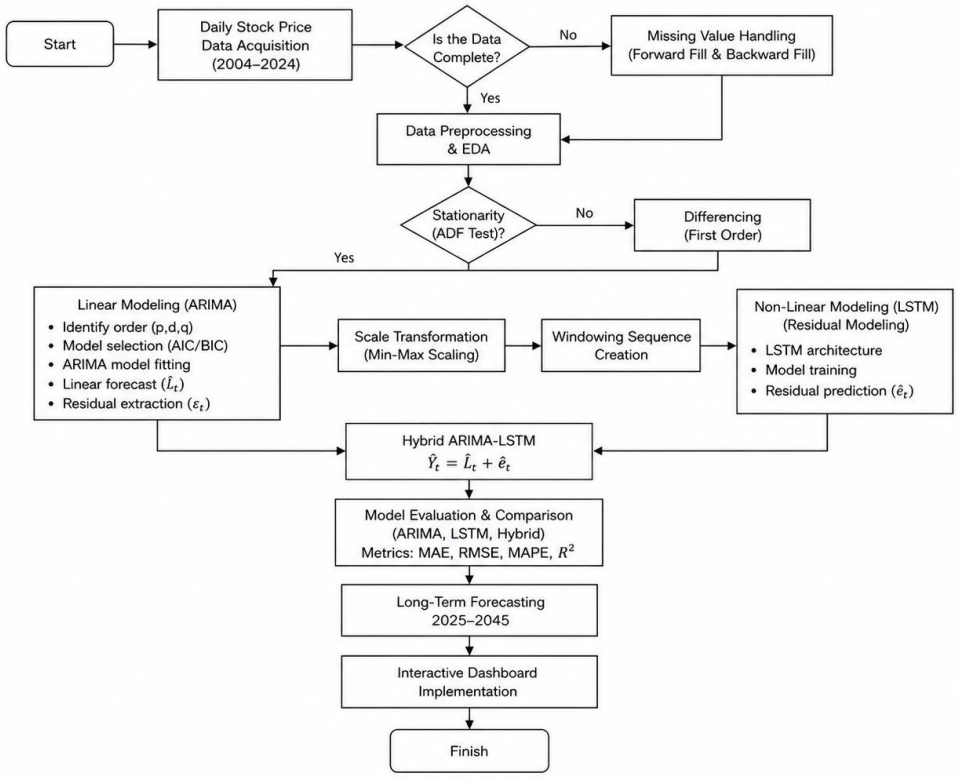


Figure 1. Research workflow of the hybrid ARIMA-LSTM model for coal-stock price forecasting.

2.2. Research Data and Dataset Split

The data consist of daily closing prices obtained from Yahoo Finance. The exact observation periods used in this study were July 16, 2008-December 30, 2024 for ADRO; January 2, 2004-December 30, 2024 for PTBA; June 11, 2008-December 30, 2024 for INDY; and December 18, 2007-December 30, 2024 for ITMG. The starting date differs because each stock has a different available trading history. The observations were sorted chronologically and examined for duplicate records and missing values. Log returns were calculated to analyze price changes, and stationarity was assessed using the Augmented Dickey-Fuller and Kwiatkowski-Phillips-Schmidt-Shin tests. Because the price series at level were non-stationary, first-order differencing was applied, resulting in an integration order of d = 1 that can be seen in equation (1).

$$\Delta P_t = P_t - P_{t-1} \quad (1)$$

The split was defined using a calendar cutoff of December 31, 2022. Accordingly, the last available training observation was December 30, 2022, while the testing period ran from January 2, 2023 to December 30, 2024. The four companies were not pooled into a single

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multivariate model. Each ticker was preprocessed, trained, and evaluated independently using its own chronological series; therefore, company-specific ARIMA, LSTM, and hybrid ARIMA-LSTM configurations were produced. This design prevents future observations from leaking into model training and allows differences in the price dynamics of each stock to be evaluated separately.

Table 1. Research dataset

Company	Ticker	Full observation period	Training period	Testing period
ADRO	ADRO.JK	16 Jul. 2008-30 Dec. 2024	16 Jul. 2008-30 Dec. 2022	2 Jan. 2023-30 Dec. 2024
PTBA	PTBA.JK	2 Jan. 2004-30 Dec. 2024	2 Jan. 2004-30 Dec. 2022	2 Jan. 2023-30 Dec. 2024
INDY	INDY.JK	11 Jun. 2008-30 Dec. 2024	11 Jun. 2008-30 Dec. 2022	2 Jan. 2023-30 Dec. 2024
ITMG	ITMG.JK	18 Dec. 2007-30 Dec. 2024	18 Dec. 2007-30 Dec. 2022	2 Jan. 2023-30 Dec. 2024

2.3. Data Preprocessing

Data preprocessing included checking and handling incomplete observations, examining unusual values, and transforming the data when necessary. For ARIMA modeling, stationarity was evaluated using the Augmented Dickey-Fuller test and confirmed with the KPSS test. Non-stationary price series were differenced to stabilize their statistical properties. The residual sequence used by the nonlinear model was transformed with Min-Max scaling and arranged into windowed input sequences.

2.4. ARIMA Model Development

ARIMA was used to capture the linear component of each stock-price series. For a price series P_t, the general ARIMA(p,d,q) formulation is expressed in Equation (2), where B denotes the backshift operator, d is the differencing order, $\varphi_p(B)$ represents the autoregressive polynomial, $\theta_q(B)$ represents the moving-average polynomial, c is a constant, and ε_t is the random error. Candidate orders were evaluated separately for each company, and the specification with the minimum Akaike Information Criterion was selected [15]. The selected configurations are summarized in Table 2.

$$\varphi_p(B)(1 - B)^d P_t = c + \theta_q(B)\varepsilon_t \quad (2)$$

The ARIMA-only output is the linear forecast produced directly by the selected company-specific ARIMA order. The forecast and its residuals are subsequently used in the hybridization stage.

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Table 2. Selected ARIMA configurations for each company

Company	Ticker	Selected ARIMA order	Model role
ADRO	ADRO.JK	ARIMA(3,1,3)	Linear price forecast
PTBA	PTBA.JK	ARIMA(1,1,3)	Linear price forecast
INDY	INDY.JK	ARIMA(1,1,3)	Linear price forecast
ITMG	ITMG.JK	ARIMA(3,1,3)	Linear price forecast

2.5. LSTM and Hybrid ARIMA-LSTM Development

The nonlinear component was implemented specifically as a Long Short-Term Memory network, which is a recurrent artificial-neural-network architecture designed for sequential data; a generic feedforward ANN was not used. For the hybrid model, the residual at time t is calculated using Equation (3), where P_t is the observed stock price and $\hat{P}_t^{ARIMA}$ is the ARIMA forecast. The residual series is Min-Max scaled and transformed into 30-day input sequences. The LSTM network then predicts the nonlinear residual component, and the final hybrid forecast is obtained using Equation (4). Thus, the ARIMA model is the linear component, the LSTM model is the residual-correction component, and the hybrid ARIMA-LSTM model is the sum of both outputs. The LSTM architecture and training settings used for every company are summarized in Table 3.

$$e_t = P_t - \hat{P}_t^{ARIMA} \tag{3}$$

The residual LSTM estimates $\hat{e}_t^{LSTM}$ from the preceding 30 residual observations. Its output is not interpreted as an independent stock price; it is a nonlinear correction that is added to the ARIMA forecast.

$$\hat{P}_t^{Hybrid} = \hat{P}_t^{ARIMA} + \hat{e}_t^{LSTM} \tag{4}$$

Model development was performed independently for the four companies. Although the same LSTM architecture was used as the initial configuration, the input data, ARIMA residuals, learned weights, and resulting forecasts differed for each ticker.

Table 3. LSTM architecture and training configuration

Component	Configuration
Input window	30 daily observations
First LSTM layer	64 units; return sequences enabled
Dropout after first LSTM	0.20

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Component	Configuration
Second LSTM layer	64 units
Dropout after second LSTM	0.20
Dense layer	32 units; ReLU activation
Output layer	1 unit; linear output
Total trainable parameters	52,033
Optimizer and learning rate	Adam; 0.001
Loss function	Mean Squared Error
Batch size and maximum epochs	32; 100 epochs
Validation and early stopping	15% validation split; patience = 15

2.6. Model Evaluation and Performance Measurement

Model performance was evaluated on the testing period using four metrics. Mean Absolute Error measures the average absolute difference between predicted and observed prices. Root Mean Squared Error penalizes larger errors more strongly. Mean Absolute Percentage Error expresses prediction error in percentage terms and supports proportional comparisons across stocks. The coefficient of determination (R^2) evaluates performance relative to a mean-based reference prediction. Lower MAE, RMSE, and MAPE values indicate smaller forecast errors, while higher R^2 values indicate stronger explanatory performance [16].

3. RESULTS AND DISCUSSION

3.1. Results

3.1.1. Data Characteristics and Stationarity

The historical data reveal substantial differences in price dynamics across the four companies. PTBA has the longest observation period, beginning in January 2004, while the ADRO, INDY, and ITMG series begin according to the availability of their respective trading records. To support comparison across stocks with different nominal prices, the historical prices were converted to indices with the initial observation of each series set to 100. A logarithmic scale was applied to improve the visibility of proportional changes.

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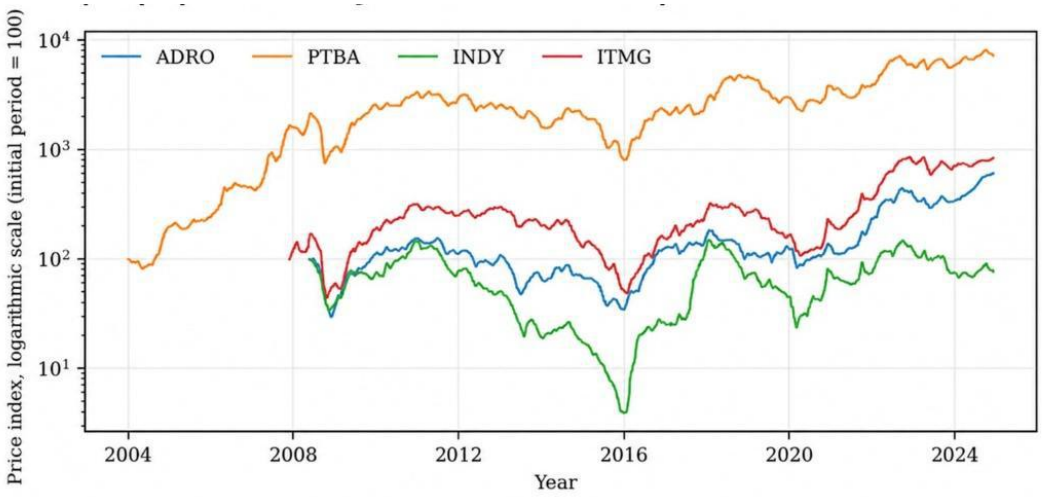


Figure 2. Indexed historical prices of Indonesian coal stocks (base = 100, logarithmic scale).

PTBA recorded the strongest cumulative increase over the observation period, although its index experienced several substantial corrections. ADRO and ITMG also exhibited long-term increases accompanied by pronounced fluctuations, while INDY showed the greatest instability, including a sharp decline around 2015-2016 and a weaker recovery thereafter. The Jarque-Bera results indicated non-normal return distributions, while the ADF and KPSS tests confirmed that the price series were non-stationary at level. After first-order differencing, the series satisfied the stationarity requirements and were modeled with d = 1 [17], [18], [19], [20], [21].

3.1.2. Model Performance During the Testing Period

To clarify how the models were formed, the ARIMA component was estimated first for each stock. The price series was differenced once because it was non-stationary at level (d = 1). Several combinations of autoregressive order p and moving-average order q were then fitted, and the specification with the lowest AIC was retained. This procedure produced ARIMA(3,1,3) for ADRO and ITMG and ARIMA(1,1,3) for PTBA and INDY. The general form of the selected ARIMA component is:

$$\varphi_p(B)(1 - B)^d P_t = c + \theta_q(B)\varepsilon_t \quad (5)$$

In this expression, the autoregressive terms use earlier values of the differenced price series, while the moving-average terms use earlier forecast errors. The resulting output, $\hat{P}_t^{ARIMA}$, is the linear stock-price forecast. After this forecast was obtained, the residual $e_t = P_t - \hat{P}_t^{ARIMA}$ was calculated and used as the LSTM input. The LSTM was not trained to predict

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the stock price independently; instead, it learned the nonlinear pattern in the previous 30 residual observations. The network consisted of two LSTM layers with 64 units, dropout of 0.20 after each layer, a 32-unit dense layer with ReLU activation, and a one-unit linear output. It was trained using Adam with a learning rate of 0.001, MSE loss, and early stopping. Its output, $\hat{e}_t^{LsTM}$, represents the nonlinear residual correction.

Accordingly, the three outputs have different roles. $\hat{P}_t^{ARIMA}$ is the ARIMA or linear forecast, $\hat{e}_t^{LsTM}$ is the LSTM or nonlinear residual correction, and $\hat{P}_t^{Hybrid} = \hat{P}_t^{ARIMA} + \hat{e}_t^{LsTM}$ is the final hybrid stock-price forecast. Table 4 reports the out-of-sample performance of the hybrid output for January 2, 2023-December 30, 2024. MAE and RMSE were not used for direct comparisons across companies because both metrics are influenced by differences in nominal stock-price scales; therefore, cross-company interpretation focused primarily on MAPE and R^2.

Table 4. Evaluation results of the hybrid ARIMA-LSTM model during the testing period

Ticker	MAE	RMSE	MAPE (%)	R^2
ADRO	151.76	170.56	11.98	0.6809
PTBA	113.55	139.95	5.29	0.5999
INDY	443.56	471.07	29.94	-1.6307
ITMG	1,657.26	1,882.08	8.50	-0.3113

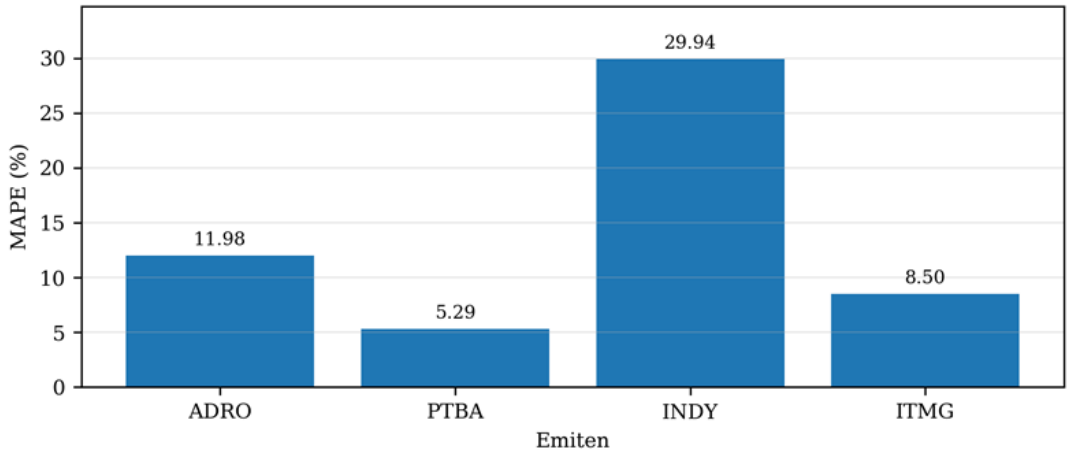


Figure 3. Comparison of MAPE values across the listed companies.

PTBA achieved the lowest MAPE of 5.29%, followed by ITMG at 8.50% and ADRO at 11.98%. INDY recorded the highest MAPE at 29.94% and an R^2 value of -1.6307. Under the adopted definition of R^2, a negative value indicates that the model produced greater squared prediction errors than a baseline model that consistently predicts the mean of the observed testing data [22].

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For PTBA, the predicted curve generally followed the direction of the actual price movements, although several peaks and declines were smoothed. This pattern is consistent with its relatively low MAPE, while the R^2 value of 0.5999 indicates that the model accounted for approximately 60% of the observed variation under the coefficient-of-determination formulation used in this study [23].

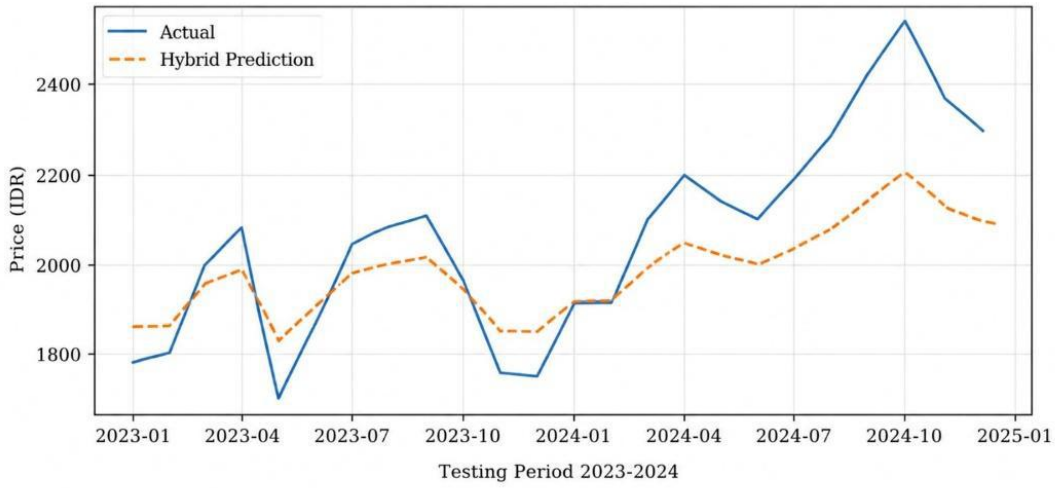


Figure 4. Actual and predicted PTBA prices during the testing period.

A more pronounced discrepancy was observed for INDY. The predicted values tended to remain above the actual prices, particularly in 2024, indicating that the model was slow to adjust to a downward shift in the price level. This persistent overestimation is consistent with the relatively high MAE and RMSE values and the negative R^2 obtained during the testing period.

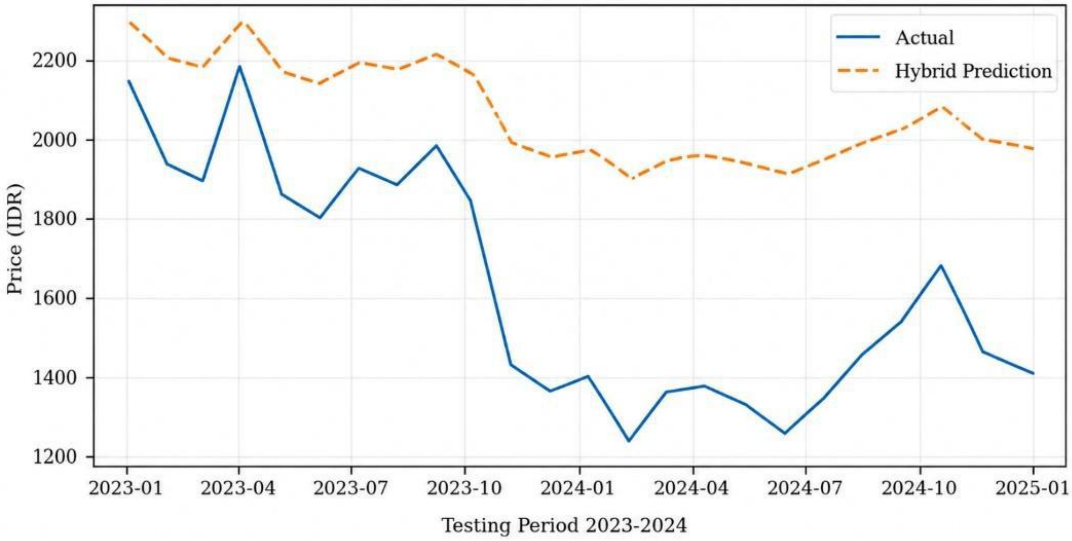


Figure 5. Actual and predicted INDY prices during the testing period.

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3.1.3. Projections for 2025-2045 and Dashboard Implementation

The selected company-specific hybrid models were used to generate monthly stock-price projections for 2025-2045. Table 5 summarizes the December 2024 prices as baseline values and the projected prices for December 2025 and December 2045. All four stocks were projected to reach higher values in 2045 than their respective positions at the end of 2024, although the projected trajectories were nonlinear and included periods of downward adjustment.

Table 5. Summary of stock-price projections for 2025-2045

Ticker	Dec. 2024	Dec. 2025	Dec. 2045	Growth (%)
ADRO	1,954.83	2,202.67	4,463.10	128.31
PTBA	2,289.88	2,556.53	4,754.61	107.64
INDY	1,411.12	1,634.66	3,029.56	114.69
ITMG	22,887.07	25,760.61	53,070.69	131.88

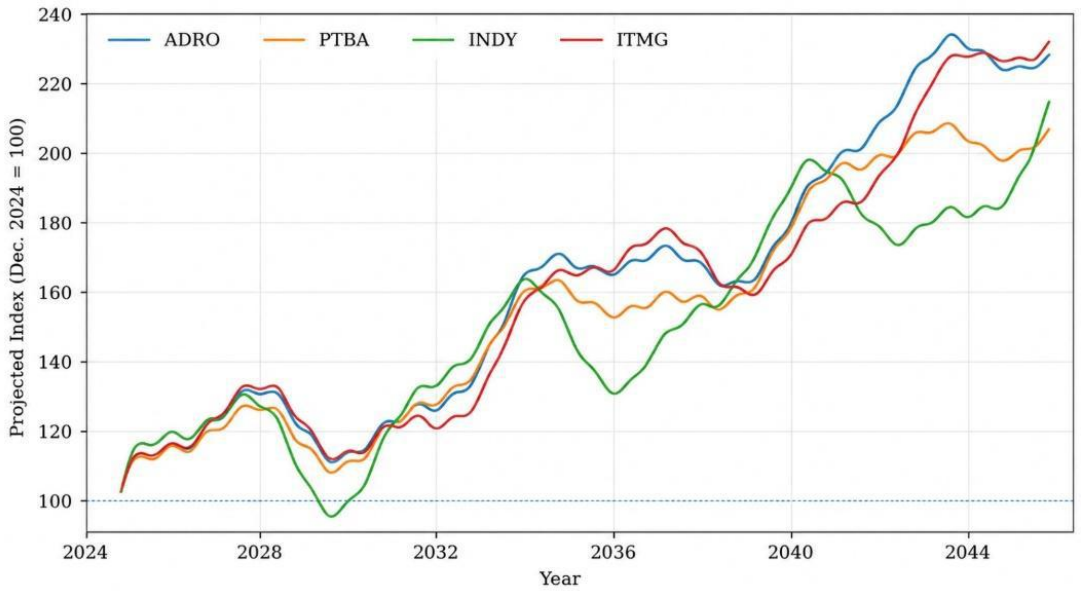


Figure 6. Stock-price projection indices for the 2025-2045 period.

Relative to December 2024, the projections through December 2045 indicate growth of approximately 128.31% for ADRO, 107.64% for PTBA, 114.69% for INDY, and 131.88% for ITMG. ITMG and ADRO show the highest relative growth, whereas PTBA exhibits a more moderate trajectory. The prediction intervals widen as the forecasting horizon increases, indicating greater uncertainty over longer horizons [24].

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The modeling results were presented through a web-based dashboard that allows users to select a stock, examine historical prices, compare actual and predicted values during the testing period, review evaluation metrics, and explore projections through 2045. The dashboard serves as a visualization and analytical medium and should not be interpreted as a stock-trading recommendation system. The interactive dashboard is publicly accessible at: https://pkm-forecast-saham-batubara-2025-20-gamma.vercel.app/

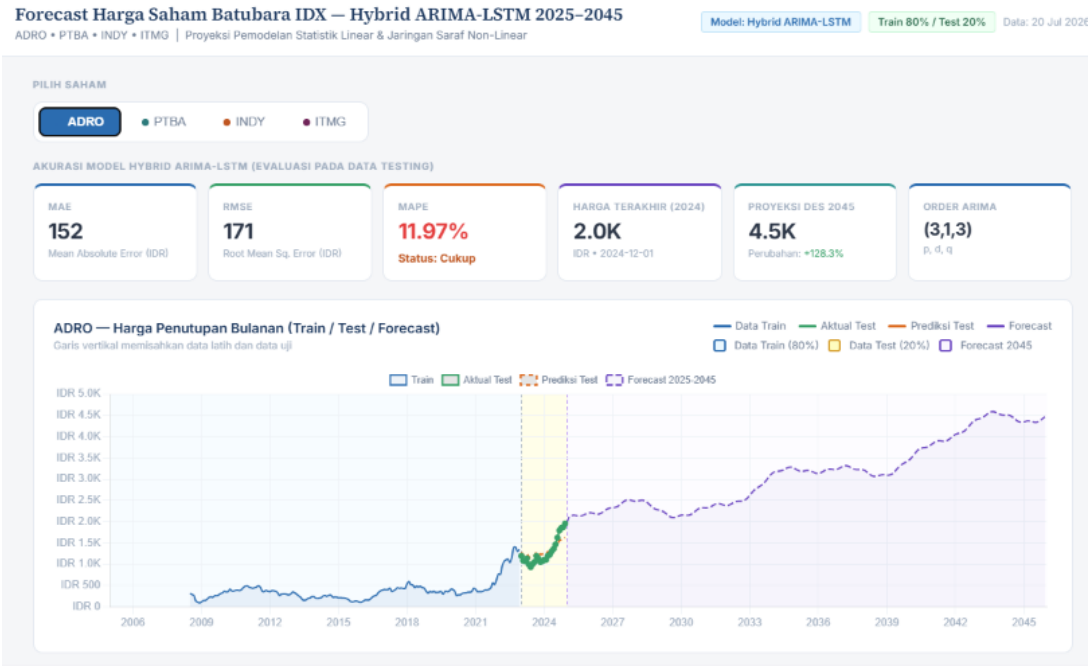
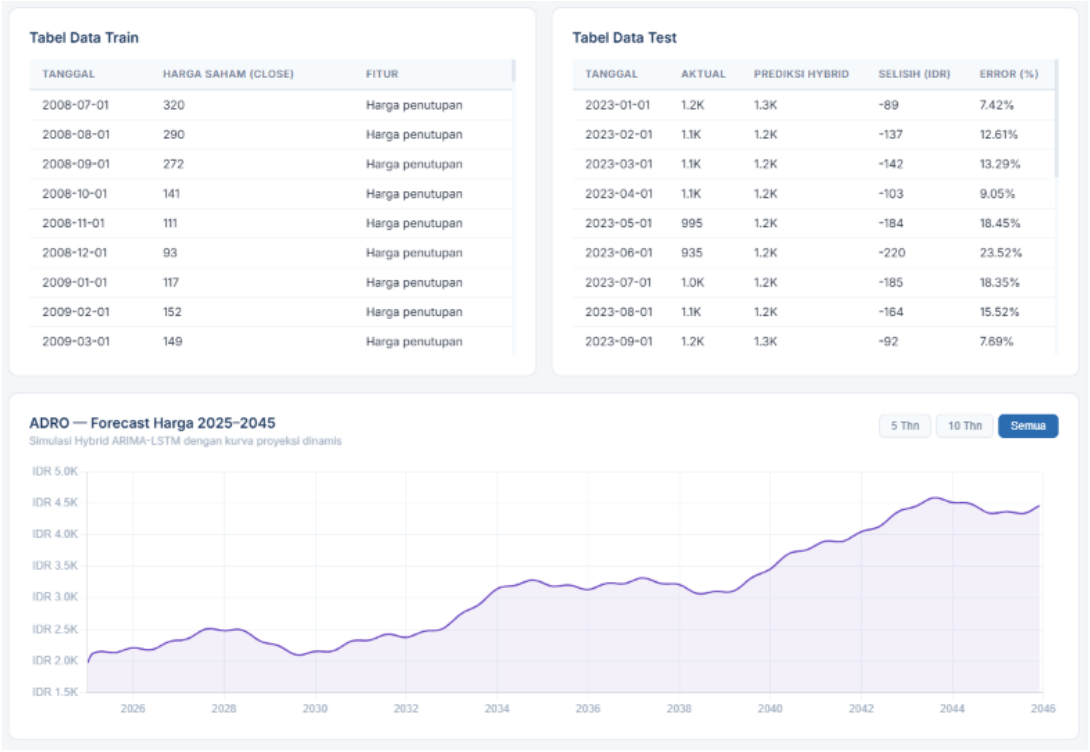


Figure 7. Main dashboard interface showing stock selection, evaluation metrics, historical data, and the 2025-2045 forecast.

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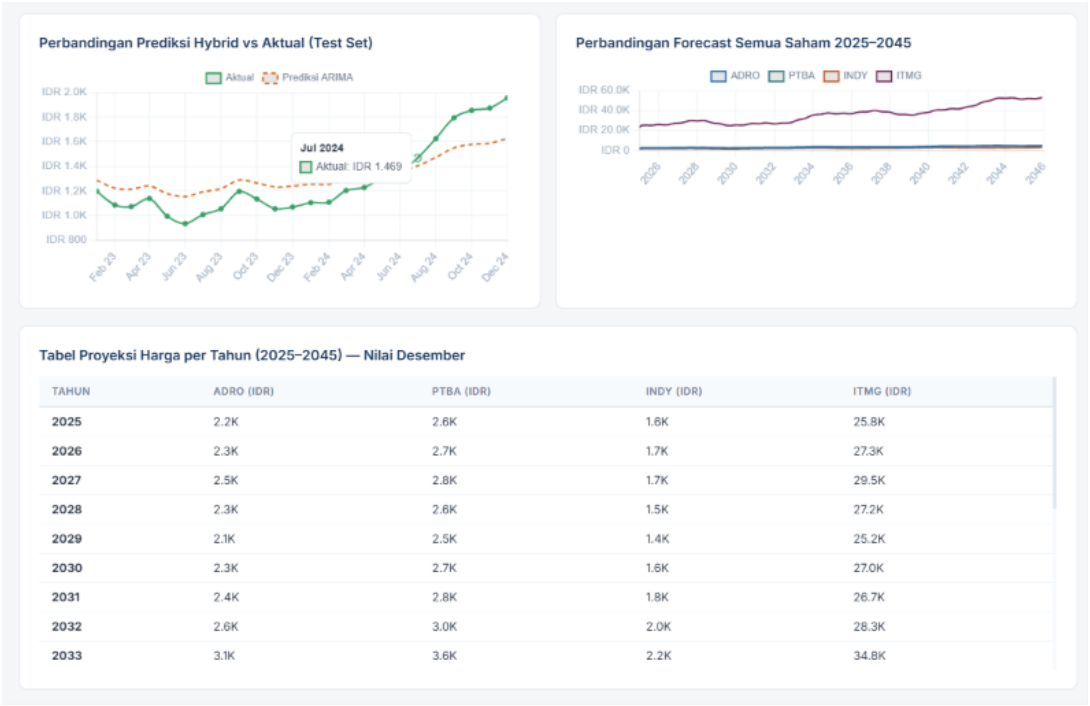
Tabel Data Train

TANGGAL	HARGA SAHAM (CLOSE)	FITUR
2008-07-01	320	Harga penutupan
2008-08-01	290	Harga penutupan
2008-09-01	272	Harga penutupan
2008-10-01	141	Harga penutupan
2008-11-01	111	Harga penutupan
2008-12-01	93	Harga penutupan
2009-01-01	117	Harga penutupan
2009-02-01	152	Harga penutupan
2009-03-01	149	Harga penutupan

Tabel Data Test

TANGGAL	AKTUAL	PREDIKSI HYBRID	SELISIH (IDR)	ERROR (%)
2023-01-01	1.2K	1.3K	-89	7.42%
2023-02-01	1.1K	1.2K	-137	12.61%
2023-03-01	1.1K	1.2K	-142	13.29%
2023-04-01	1.1K	1.2K	-103	9.05%
2023-05-01	995	1.2K	-184	18.45%
2023-06-01	935	1.2K	-220	23.52%
2023-07-01	1.0K	1.2K	-185	18.35%
2023-08-01	1.1K	1.2K	-164	15.52%
2023-09-01	1.2K	1.3K	-92	7.69%

Figure 8. Training and testing data tables and the detailed stock-specific forecast for 2025-2045.



Tabel Proyeksi Harga per Tahun (2025–2045) — Nilai Desember

TAHUN	ADRO (IDR)	PTBA (IDR)	INDY (IDR)	ITMG (IDR)
2025	2.2K	2.6K	1.6K	25.8K
2026	2.3K	2.7K	1.7K	27.3K
2027	2.5K	2.8K	1.7K	29.5K
2028	2.3K	2.6K	1.5K	27.2K
2029	2.1K	2.5K	1.4K	25.2K
2030	2.3K	2.7K	1.6K	27.0K
2031	2.4K	2.8K	1.8K	26.7K
2032	2.6K	3.0K	2.0K	28.3K
2033	3.1K	3.6K	2.2K	34.8K

Figure 9. Actual-versus-predicted comparison, multi-stock forecast visualization, and annual projection summary.

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3.2. Discussion

The results show that the hybrid approach did not produce equal predictive performance across all stocks. The ARIMA component was designed to capture linear dependencies, whereas the LSTM component modeled nonlinear patterns remaining in the ARIMA residuals [25]. Nevertheless, the nonlinear correction did not consistently improve forecasts when substantial shifts in price levels occurred, particularly for INDY. The superiority of a hybrid model should therefore not be assumed automatically and must be established through stock-specific out-of-sample evaluation.

Differences in performance are consistent with the distinct price trajectories and volatility patterns observed across the companies. Structural changes may alter model parameters and forecast errors, causing a configuration that performs well for one stock to perform poorly for another [26]. Rolling-origin validation would provide a stronger assessment because it evaluates performance across multiple forecast origins rather than relying on a single testing period [27]. Stock-specific optimization of window length, network architecture, and training parameters should also be considered.

External predictors may improve the model's responsiveness to changing market conditions. Benchmark coal prices, exchange rates, market indices, inflation, interest rates, trading volume, and export policies are potentially relevant because multivariate forecasting models can integrate technical, macroeconomic, and fundamental information [28], [29]. Their usefulness must still be tested empirically because structural breaks can weaken relationships that were reliable in earlier periods [30].

The long-term projections should be interpreted as exploratory scenarios rather than definite price targets. Forecast uncertainty becomes increasingly difficult to estimate as the prediction horizon expands [31]. Consequently, the dashboard is most appropriately used as a market-monitoring and analytical tool. Future development should include time-series cross-validation, periodic model updating, stock-specific hyperparameter optimization, and uncertainty intervals calibrated specifically for the hybrid model [32].

4. CONCLUSION

This study developed a coal-sector stock-price forecasting framework integrating ARIMA, LSTM, and hybrid ARIMA-LSTM models, with the results presented through an interactive dashboard. Evaluation for January 2023-December 2024 showed different performance across the selected stocks, with MAPE values of 5.29% for PTBA, 8.50% for ITMG, 11.98% for ADRO, and 29.94% for INDY. Positive R^2 values for ADRO and PTBA indicate that the model explained part of the observed price variation, whereas negative values for ITMG and INDY show that their dynamics were not adequately represented by the same configuration. These findings demonstrate that adding an LSTM component to ARIMA residuals does not automatically yield uniform improvements and that model selection should be based on the characteristics and evaluation results of each company. The selected models generated exploratory monthly projections for 2025-2045 and the dashboard improved the accessibility of historical, evaluation, and projection information. The main practical impact is an analytical interface for monitoring coal-sector price scenarios, while the principal methodological implication is the need for stock-specific validation, periodic updating, external predictors, and calibrated uncertainty estimates.

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REFERENCES

[1] A. N. Silvia Dan N. Aslami, "Analisis Kebijakan Perdagangan Internasional," *J. Econ. Curr. Study*, Vol. 4, 2022.

[2] N. Hanif Dan M. Taufiq, "Pengaruh Nilai Tukar, Volume Produksi, Hba, dan Harga Minyak Dunia Terhadap Nilai Ekspor Batubara Indonesia," *J. Ekon. Pembang. Stie Muhammadiyah Palopo*, Vol. 9, No. 1, Hal. 267, 2023, Doi: Https://Doi.Org/10.35906/Jep.V9i1.1512.

[3] P. Astuty Dan A. Raymond Trilaksana, "Contribution Of Coal Export Performance

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to Gross Regional Domestic Product and Non-Tax State Revenue In The Kalimantan and Sumatera Regions," *Int. J. Eng. Bus. Soc. Sci.*, Vol. 2, No. 03, Hal. 1060–1066, Jan 2024, Doi: 10.58451/Ijebss.V2i03.129.

[4] S. Ayu Arsita, G. Eko Saputro, Dan S. Susanto, "Perkembangan Kebijakan Energi Nasional dan Energi Baru Terbarukan Indonesia," *J. Syntax Transform.*, Vol. 2, No. 12, Hal. 1779–1788, Des 2021, Doi: 10.46799/Jst.V2i12.473.

[5] M. Fadli, A. H. Jaya, Dan Y. Kornelius, "Strategi Bauran Pemasaran Briket Pengganti Arang Di Desa Sibalaya Selatan," *J. Pengabdi. Pada Masy. Indones.*, Vol. 2, No. 2, Hal. 70–77, Apr 2023, Doi: 10.55542/Jppmi.V2i2.619.

[6] N. Wati Dan A. Puspitaningtyas, "Pengaruh Volume Perdagangan Saham, Tingkat Inflasi dan Nilai Tukar Terhadap Volatilitas Harga Saham Sub Sektor Batu Bara Yang Terdaftar Di Bursa Efek Indonesia," Vol. 11, No. 2, 2023.

[7] L. Rulíšek, "Relationship Between Palm Oil And Coal Prices On The Growth Of The Trade Balance In 2009 - 2020," 20 Juni 2022. Doi: 10.31219/Osf.Io/6db7q.

[8] R. Pahlevi, S. Thamrin, I. Ahmad, dan F. B. Nugroho, "Masa Depan Pemanfaatan Batubara Sebagai Sumber Energi Di Indonesia," *J. Energi Baru Dan Terbarukan*, Vol. 5, No. 2, Hal. 50–60, 2024, Doi: Https://Doi.Org/10.14710/Jebt.2024.22973.

[9] I. Nuryanneti, S. Amina, dan P. Putra, "Peningkatan Kualitas Briket Melalui Pemanfaatan Limbah Fine Coal Dan Biomassa: Studi Di Pabrik Briket Tanjung Enim Pt. Bukit Asam, Tbk.," *J. Innov. Creat.*, Vol. 5, No. 3, 2025, Doi: Https://Doi.Org/10.31004/Joecy.V5i3.4075.

[10] T. Di. J. M. Dan B. K. E. Dan S. D. Mineral, *Road Map Pengembangan Dan Pemanfaatan Batubara*. Jakarta: Direktorat Jenderal Mineral Dan Batubara, 2021.

[11] R. G. Ahangar, M. Yahyazadehfar, Dan H. Pournaghshband, "The Comparison Of Methods Artificial Neural Network With Linear Regression Using Specific Variables For Prediction Stock Price In Tehran Stock Exchange," *Int. J. Comput. Sci. Inf. Secur.*, Vol. 7, No. 2, Hal. 38–46, 2010, Doi: Https://Doi.Org/10.48550/Arxiv.1003.1457.

[12] A. Alamsyah Dan M. F. Permana, "Artificial Neural Network For Predicting Indonesian Economic Growth Using Macroeconomics Indicators," *Ieee*, Vol. 1, No. 1, Hal. 15–19, 2018, Doi: Https://Doi.Org/10.1109/Sain.2018.8673347.

[13] S. A. A. Kharis, A. H. A. Zili, M. Malik, W. Nuryaningrum, Dan A. Putri, "Comparing Machine Learning Models For Indonesia Stock Market Prediction," *Indones. J. Electr. Eng. Comput. Sci.*, Vol. 38, No. 1, Hal. 508–516, 2025, Doi:

Http://Doi.Org/10.11591/Ijeecs.V38.I1.Pp508-516.

[14] L. Slater *Et Al.*, "Hybrid Forecasting Using Statistics And Machine Learning To Integrate Prediction From Dynamical Models," *Hydrol. Earth Syst. Sci.*, Vol. 1, No. 1, Hal. 1–35, 2022.

[15] S. Elsworth Dan S. Güttel, "Time Series Forecasting Using Lstm Networks: A Symbolic Approach," *Arxiv Prepr. Arxiv2003.05672*, 2020.

[16] L. S. Ihzaniah, A. Setiawan, Dan R. W. N. Wijaya, "Perbandingan Kinerja Metode Regresi K-Nearest Neighbor Dan Metode Regresi Linear Berganda Pada Data Boston Housing," *Jambura J. Probab. Stat.*, Vol. 4, No. 1, Hal. 17–29, 2023.

[17] C. M. Jarque Dan A. K. Bera, "Efficient Tests For Normality, Homoscedasticity And Serial Independence Of Regression Residuals," *Econ. Lett.*, Vol. 6, No. 3, Hal. 255–259, 1980, Doi: Https://Doi.Org/10.1016/0165-1765(80)90024-5.

[18] S. E. Said Dan D. A. Dickey, "Testing For Unit Roots In Autoregressive-Moving Average Models Of Unknown Order," *Biometrika*, Vol. 71, No. 3, Hal. 599–607, 1984, Doi: 10.1093/Biomet/71.3.599.

[19] D. Kwiatkowski, P. C. B. Phillips, P. Schmidt, Dan Y. Shin, "Testing The Null Hypothesis Of Stationarity Against The Alternative Of A Unit Root: How Sure Are We That Economic Time Series Have A Unit Root?," *J. Econom.*, Vol. 54, No. 1, Hal. 159–178, 1992, Doi: Https://Doi.Org/10.1016/0304-4076(92)90104-Y.

[20] R. J. Hyndman Dan Y. Khandakar, "Automatic Time Series Forecasting: The Forecast Package For R," *J. Stat. Softw.*, Vol. 27, No. 3, Hal. 1–22, 2008, Doi: 10.18637/Jss.V027.I03.

[21] C. W. J. Granger Dan P. Newbold, "Spurious Regressions In Econometrics," *J. Econom.*, Vol. 2, No. 2, Hal. 111–120, 1974, Doi: Https://Doi.Org/10.1016/0304-4076(74)90034-7.

[22] D. Chicco, M. Warrens, Dan G. Jurman, "The Coefficient Of Determination R-Squared Is More Informative Than Smape, Mae, Mape, Mse And Rmse In Regression Analysis Evaluation," *Peerj Comput. Sci.*, Vol. 7, No. E623, 2021.

[23] T. O. Kvålseth, "Cautionary Note About R 2," *Am. Stat.*, Vol. 39, No. 4, Hal. 279–285, 1985, Doi: 10.1080/00031305.1985.10479448.

[24] M. Chudý, S. Karmakar, Dan W. B. Wu, "Long-Term Prediction Intervals Of Economic Time Series," *Empir. Econ.*, Vol. 58, No. 1, Hal. 191–222, 2020, Doi: 10.1007/S00181-019-01689-2.

[25] G. P. Zhang, "Time Series Forecasting Using A Hybrid Arima And Neural Network

Model," *Neurocomputing*, Vol. 50, Hal. 159–175, 2003, Doi: Https://Doi.Org/10.1016/S0925-2312(01)00702-0.

[26] M. H. Pesaran Dan A. Timmermann, "Small Sample Properties Of Forecasts From Autoregressive Models Under Structural Breaks," *J. Econom.*, Vol. 129, No. 1, Hal. 183–217, 2005, Doi: Https://Doi.Org/10.1016/J.Jeconom.2004.09.007.

[27] L. J. Tashman, "Out-Of-Sample Tests Of Forecasting Accuracy: An Analysis And Review," *Int. J. Forecast.*, Vol. 16, No. 4, Hal. 437–450, 2000, Doi: Https://Doi.Org/10.1016/S0169-2070(00)00065-0.

[28] H. N. Bhandari, B. Rimal, N. R. Pokhrel, R. Rimal, K. R. Dahal, Dan R. K. C. Khatri, "Predicting Stock Market Index Using Lstm," *Mach. Learn. With Appl.*, Vol. 9, Hal. 100320, 2022, Doi: Https://Doi.Org/10.1016/J.Mlwa.2022.100320.

[29] M. M. Kumbure, C. Lohrmann, P. Luukka, Dan J. Porras, "Machine Learning Techniques And Data For Stock Market Forecasting: A Literature Review," *Expert Syst. Appl.*, Vol. 197, Hal. 116659, 2022, Doi: Https://Doi.Org/10.1016/J.Eswa.2022.116659.

[30] J. L. Castle, M. P. Clements, Dan D. F. Hendry, "An Overview Of Forecasting Facing Breaks," *J. Bus. Cycle Res.*, Vol. 12, No. 1, Hal. 3–23, 2016, Doi: 10.1007/S41549-016-0005-2.

[31] M. Knüppel, "Forecast-Error-Based Estimation Of Forecast Uncertainty When The Horizon Is Increased," *Int. J. Forecast.*, Vol. 34, No. 1, Hal. 105–116, 2018, Doi: Https://Doi.Org/10.1016/J.Ijforecast.2017.08.006.

[32] C. Bergmeir Dan J. M. Benítez, "On The Use Of Cross-Validation For Time Series Predictor Evaluation," *Inf. Sci. (Ny).*, Vol. 191, Hal. 192–213, 2012, Doi: Https://Doi.Org/10.1016/J.Ins.2011.12.028.